Solution Sets of Geometric Hybrid Dynamical Systems: Topological Properties and Existence of Extremal Solutions

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***Abstract*— We present a topological perspective to guarantee structural properties of the sets of solutions to geometric hybrid systems defined on a finite-dimensional manifold. By identifying a solution with its graph, we study maximal solutions as points in the hyperspace of closed nonempty subsets of the product of the hybrid-time space with the manifold. We equip this hyperspace with the Fell topology, and show that sequential compactness and semicontinuous dependence on initial conditions of the solution map correspond to topological compactness and outer semicontinuity in the Fell topology. This equivalence enables the application of theorems on the existence of extremal solutions for cost functionals as well as measurable selection of such solutions when initial conditions are drawn from a measurable space.**

I. Introduction

The hybrid inclusions framework [1] models dynamical systems that combine continuous-time and discrete-time evolution, by using set-valued flow and jump maps, together with constraint sets specifying where each type of evolution is admissible. In this framework, solutions need not be unique, which accommodates a broad class of physical systems [2, Ch. 2]. While nonuniqueness of solutions enables generality in the modeling of systems, it also makes more difficult the study of the structure of the associated solution sets. The hybrid inclusions framework addresses this challenge by using nominal well-posedness, a property requiring that each element obtained as a suitable limit of solutions is itself a solution. Among others, this closure property ensures that solution sets depend semicontinuously on initial conditions, that forward reachable sets are compact, and that sufficient regularity of omega limit sets of solutions is available for invariance principles and stability analysis [3].

The existing theory of hybrid inclusions in the Euclidean setting uses sequential characterizations to analyze these properties of solution sets. For instance, compactness of solution sets and semicontinuous dependence of solutions on initial conditions are formulated via Painlevé–Kuratowski convergence of graphs of solutions, referred to as graphical convergence [1, Def. 5.21],[4]. This sequential approach has been effective for developing invariance principles and robust stability theory. However, results in variational analysis relevant to worst-case analysis of systems with nonunique solutions usually require topological characterizations. For instance, to obtain regularity of the set of minimizers of cost functionals defined on the values of maps, [5, Thm. 18.19] requires the associated maps to be outer semicontinuous and compact valued in a given topology. To the best of our knowledge, there is no readily available result that asserts the existence of such topology for the solution map of geometric hybrid systems modeled as hybrid inclusions, or if existing sequential properties are sufficient to identify it.

In the Euclidean setting, these questions can be addressed by representing each maximal solution by its graph, viewed as a point in the space of closed nonempty subsets of $\mathbb{R}_{\geq 0} \times \mathbb{N} \times \mathbb{R}^n$, and metrizing this space using the integrated set distance [6, Eq. 4(12)]; see also [7]. The topology that this distance induces coincides with Painlevé-Kuratowski convergence [6, Thm. 4.4.2], thus allowing a connection between the sequential and topological characterizations of the solution map. This construction, however, relies on the ambient Euclidean metric, and, to date, it is unknown if it extends to hybrid dynamical systems on C^1-manifolds [3], [8]. In such setting, the state space does not have a canonical metric available since logical states may introduce disconnected components. Bridging those components metrically requires a choice of points in the space, through which distances can be defined between different components; see [9, Cor. 13.30]. Any such choice imposes additional structure with, in general, no intrinsic relationship to the dynamics. These limitations raise the question of whether an intrinsically topological approach can recover and extend the outer semicontinuity and compact valuedness of the solution map to the geometric setting.

In this paper, to address the above question,

- (i) we show that, under nominal well-posedness of a geometric hybrid system defined on a C^1 manifold $\mathcal{M}$, maximal solutions have closed graph, which allows their identification with points in the hyperspace of closed and nonempty subsets of $\mathbb{R}_{\geq 0} \times \mathbb{N} \times \mathcal{M}$;
- (ii) under pre-forward completeness of a nominally well-posed geometric hybrid system from a neighborhood of a set $\Lambda \subset \mathcal{M}$, we establish outer semicontinuity and local precompactness of the solution map with respect to the Fell topology on such hyperspace;
- (iii) we derive measurability of the solution map with respect to the Borel σ-algebra generated by the Fell topology;
- (iv) we establish existence of extremal solutions for cost functionals in the hyperspace, and measurable selection of maximizers when initial conditions are drawn from a measurable space.

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II. Preliminaries

A. Notation

The set of real numbers and the set of natural numbers, including 0, are denoted by $\mathbb{R}$ and $\mathbb{N}$, respectively. To simplify notation, for two (or more) vectors $u, v \in \mathbb{R}^n$, we write (u, v) to denote their concatenation. The *composition of two maps* $f : X \to Y$ and $g : Y \to Z$ is denoted by $g \circ f : X \to Z$. A *set-valued map* $\Gamma : X \rightrightarrows Y$ assigns to each $x \in X$ a subset $\Gamma(x) \subset Y$. The *domain, range, and graph* of Γ are defined as $\operatorname{dom}\Gamma := \{x : \Gamma(x) \neq \varnothing\}$, $\operatorname{rge}\Gamma := \{y : \exists x \text{ s.t. } y \in \Gamma(x)\}$, and $\operatorname{gph}\Gamma := \{(x, y) : y \in \Gamma(x)\}$, respectively. The *restriction of* Γ *to a set* $A \subset X$ is the set-valued map $\Gamma|_A : X \rightrightarrows Y$ defined by $\Gamma|_A(x) := \Gamma(x)$ for $x \in A$ and $\Gamma|_A(x) := \varnothing$ for $x \notin A$. Given $B \subset Y$, the *inverse image* of B under Γ is defined by $\Gamma^{-1}(B) := \{x \in X : \Gamma(x) \cap B \neq \varnothing\}$.

B. Topology

A topological space consists of a set X together with a collection $\mathcal{T}$ of subsets of X, called a *topology* on X. The elements of $\mathcal{T}$ are called *open sets*, and a set is *closed* if its complement is open. For a set $A \subset X$, the *closure* of A, denoted by $\overline{A}$, is the intersection of all closed sets containing A, and the *interior* of A, denoted by $\operatorname{int} A$, is the union of all open sets contained in A. A set A is *compact* if every open cover has a finite subcover, and *precompact* if $\overline{A}$ is compact. Given a set $A \subset X$, the set of *neighborhoods* of A is denoted by $\mathcal{N}(A) := \{U \subset X : \exists V \in \mathcal{T}, A \subset V \subset U\}$; when $A = \{x\}$ for $x \in X$, we write $\mathcal{N}(x)$ instead of $\mathcal{N}(\{x\})$. Given a subset $Y \subset X$, the *subspace topology* on Y is defined by $\mathcal{T}_Y := \{Y \cap U : U \in \mathcal{T}\}$.

A topological space $(X, \mathcal{T})$ is *Hausdorff* if, for each $x \neq y$, there exist disjoint sets $U, V \in \mathcal{T}$ with $x \in U$ and $y \in V$. A *base* for the topology $\mathcal{T}$ is a collection $\mathcal{B}$ of subsets of X such that every set in $\mathcal{T}$ can be written as a union of sets from $\mathcal{B}$. A *neighborhood base* at x, denoted $\mathcal{B}(x)$, is a collection of neighborhoods of x such that for every neighborhood $\mathcal{U} \in \mathcal{N}(x)$ there exists $B \in \mathcal{B}(x)$ with $B \subset \mathcal{U}$. A *subbase* for the topology $\mathcal{T}$ is a collection $\mathcal{S}$ of subsets of X such that the collection of all finite intersections of sets from $\mathcal{S}$ forms a base for $\mathcal{T}$. The topological space $(X, \mathcal{T})$ is *first countable* if every point has a countable neighborhood base, *second countable* if the topology admits a countable base, and *separable* if there is a countable subset $Y \subset X$ with $\overline{Y} = X$. The space is *locally compact* if every $x \in X$ has a compact neighborhood. The space is *metrizable* if there exists a metric $d : X \times X \to \mathbb{R}_{\geq 0}$ such that $\mathcal{T}$ coincides with the topology generated by d. The space is *completely metrizable* if it is metrizable by a complete metric.

An n-dimensional C^1*-manifold* $\mathcal{M}$ is a Hausdorff, second countable topological space together with a C^1-atlas [10, Def. 8.2]. The *tangent space* to $\mathcal{M}$ at $x \in \mathcal{M}$ is denoted by $T_x\mathcal{M}$, and the tangent bundle by $T\mathcal{M}$. The tangent map $\mathrm{d}f_x$ of f at x is defined by $\mathrm{d}f_x(v) := \frac{d}{dt}(f \circ \gamma)(t)\big|_{t=0}$ for each $v \in T_x\mathcal{M}$, where $\gamma : \mathbb{R} \supset I \to \mathcal{M}$ satisfies $\gamma(0) = x$ and $\frac{d\gamma}{dt}(0) = v$. For details on differential geometry and general topology, we refer the reader to [9], [11].

C. Set Convergence and Anatomy of Set-Valued Maps

Set-valued maps appear throughout the analysis of hybrid systems, both in the system data, where flow and jump maps are set valued, and in the solution map that assigns to each initial condition a set of solutions. To analyze these maps, we review sequential convergence of sets

Definition 2.1 (Painlevé-Kuratowski Convergence [12, Prop. 5.2.2])**.** *Let X be a topological space and $\{A_k\}_{k\in\mathbb{N}}$ be a sequence of subsets of X. The inner and outer limit of the sequence are defined, respectively, as*

$$\liminf_{k\to\infty} A_k := \{x \in X : \forall U \in \mathcal{N}(x), \exists k_0 \in \mathbb{N} \text{ s.t. } U \cap A_k \neq \varnothing \ \forall k \geq k_0\}$$

$$\limsup_{k\to\infty} A_k := \{x \in X : \forall U \in \mathcal{N}(x), \forall k_0 \in \mathbb{N}, \exists k \geq k_0 \text{ s.t. } U \cap A_k \neq \varnothing\}$$

The sequence converges in the Painlevé-Kuratowski sense to a set $A \subset X$ if $\limsup_{k\to\infty} A_k = \liminf_{k\to\infty} A_k = A$, in which case we write $\lim_{k\to\infty} A_k = A$ or $A_k \to A$.

Using Painlevé-Kuratowski convergence, we can formalize the following regularity property of set-valued maps along converging sequences of points.

Definition 2.2 (Outer Semicontinuous Maps [12, Def. 6.1.14])**.** *Let X and Y be first countable Hausdorff spaces and $\Gamma : X \rightrightarrows Y$ be a set-valued map. The map Γ is outer semicontinuous at $x_0 \in X$ if, for every sequence $\{x_k\}_{k\in\mathbb{N}} \to x_0$, we have $\limsup_{k\to\infty} \Gamma(x_k) \subset \Gamma(x_0)$. The map Γ is said to be outer semicontinuous if it is outer semicontinuous at every $x_0 \in X$.*

Local precompactness of set-valued maps requires images of neighborhoods to have compact closure.

Definition 2.3 (Locally Precompact Maps)**.** *A set-valued map $\Gamma : X \rightrightarrows Y$ between Hausdorff spaces is said to be locally precompact at $x \in X$ if there exists an open neighborhood U of x with $\overline{\Gamma(U)}$ compact.*

Measurability of set-valued maps captures the compatibility of their preimages with a given σ-algebra[1], and enables the extraction of measurable selections from the map. For a topological space $(X, \mathcal{T})$, the *Borel σ-algebra* $\mathfrak{B}(X)$ is the σ-algebra generated by the collection of sets defining $\mathcal{T}$.

Definition 2.4 (Measurable Maps [5, Def. 18.1])**.** *Let $(X, \mathfrak{A})$ be a measurable space, Y be a Hausdorff space, and $\Gamma : X \rightrightarrows Y$ be a set-valued map. The map Γ is said to be*

- *weakly measurable if $\Gamma^{-1}(U) \in \mathfrak{A}$ for every open set U;*
- *measurable if $\Gamma^{-1}(S) \in \mathfrak{A}$ for every closed set S.*

When Y is a separable metrizable space and Γ is compact valued, these two notions coincide [5, Lem. 18.2]. In such a case, we refer to Γ as a measurable set-valued map.

[1] A *σ-algebra* on a set X is a collection $\mathfrak{A}$ of subsets of X that contains $\varnothing$, is closed under complements, and is closed under countable unions. The pair $(X, \mathfrak{A})$ is called a *measurable space*.

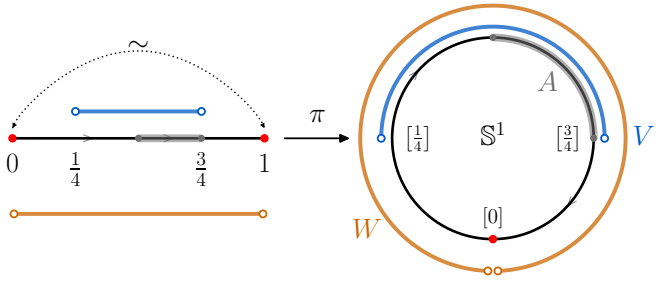


Fig. 1: The construction in Example 3.4. The circle $\mathbb{S}^1$ is the quotient of $[0,1]$ identifying 0 with 1, and π is the associated projection. Each set on the left projects to the set of the same color on the right. The closed set A is drawn in gray. The open set V, drawn in blue, intersects A. The open set W, drawn in orange, is the complement of the compact set $\{[0]\}$ and contains A. White circles mark the ends of open sets.

III. Hyperspaces and the Fell Topology

A *hyperspace* of a topological space $(X, \mathcal{T})$ is a space whose points are subsets of X, equipped with a topology reflecting geometric relationships between those subsets. We present the Fell topology on the hyperspace of closed nonempty subsets of a Hausdorff space X, denoted $\mathrm{CL}_{\neq\varnothing}(X)$, and its equivalence to Painlevé-Kuratowski convergence under appropriate regularity conditions. We begin by introducing two families of subsets of $\mathrm{CL}_{\neq\varnothing}(X)$ that serve as building blocks for the Fell topology.

Definition 3.1 (Miss and Hit Sets [12, Section 2.2])**.** *Given a set $A \subset X$, let $A^+ := \{S \in \mathrm{CL}_{\neq\varnothing}(X) : S \subset A\}$ and let $A^- := \{S \in \mathrm{CL}_{\neq\varnothing}(X) : S \cap A \neq \varnothing\}$. The set A^+ consists of the sets that "miss" the complement of A, whereas the set A^- consists of closed sets that "hit" A.*

Example 3.2. *Consider $X = \mathbb{R}$ and let $A = [0,1]$. The set A^+ contains all closed sets contained in $[0,1]$, such as $\{1/2\}$ or $[1/4, 3/4]$, but excludes $[0,2]$ since it extends outside $[0,1]$. The set A^- contains all closed sets in X that intersect the interval $[0,1]$, such as $[0,2]$, $\{1/2\}$, or $[-1,\infty)$.*

The following definition formalizes the Fell topology using these two families of sets.

Definition 3.3 (Fell Topology, [12, Def. 5.1.1])**.** *Given a Hausdorff space $(X, \mathcal{T})$, the Fell topology on $\mathrm{CL}_{\neq\varnothing}(X)$, denoted $\mathcal{T}_\mathcal{F}$, is the topology generated by the subbase*

$$\mathcal{S}_\mathcal{F} = \{V^- : V \text{ nonempty open in } \mathcal{T}\} \cup \{U^+ : U \text{ nonempty open in } \mathcal{T}, X \setminus U \text{ compact in } \mathcal{T}\},$$

where the union is taken in the hyperspace $\mathrm{CL}_{\neq\varnothing}(X)$. We write $(\mathrm{CL}_{\neq\varnothing}(X), \mathcal{T}_\mathcal{F})$ to denote the hyperspace $\mathrm{CL}_{\neq\varnothing}(X)$ with the Fell topology, and $A_k \xrightarrow{\mathcal{T}_\mathcal{F}} A$ for convergence[2] of a sequence $\{A_k\}_{k\in\mathbb{N}} \subset \mathrm{CL}_{\neq\varnothing}(X)$ to $A \in \mathrm{CL}_{\neq\varnothing}(X)$ in $\mathcal{T}_\mathcal{F}$.

Example 3.4 (An Open Neighborhood in $(\mathrm{CL}_{\neq\varnothing}(\mathbb{S}^1), \mathcal{T}_\mathcal{F})$)**.** *Consider the circle $\mathbb{S}^1 = [0,1]/\!\sim$, where $\sim$ is the equivalence relation on $[0,1]$ defined by $0 \sim 1$ and $x \sim x$ for all $x \in [0,1]$. Equip $[0,1]$ with the metric topology induced by the absolute value $|\cdot|$, and $\mathbb{S}^1$ with the quotient topology[3] $\mathcal{T}_\pi$ induced by the projection $\pi : [0,1] \to \mathbb{S}^1$ that maps each $x \in [0,1]$ to its equivalence class $[x] \in \mathbb{S}^1$. The space $(\mathbb{S}^1, \mathcal{T}_\pi)$ is Hausdorff. Now, let $A := \pi([1/2, 3/4]) \in \mathrm{CL}_{\neq\varnothing}(\mathbb{S}^1)$, $V := \pi((1/4, 3/4))$, and $W := \mathbb{S}^1 \setminus \{[0]\}$; this construction is illustrated in Figure 1. By Definition 3.3, V^- belongs to the subbase defining $\mathcal{T}_\mathcal{F}$ since V is nonempty and open. Similarly, W^+ belongs to the subbase since W is open with compact complement $\{[0]\}$. Since finite intersections of subbase elements are open, $U := V^- \cap W^+$ is open in $(\mathrm{CL}_{\neq\varnothing}(\mathbb{S}^1), \mathcal{T}_\mathcal{F})$.*

Givent that $A \cap V = \pi([1/2, 3/4)) \neq \varnothing$, we have that $A \in V^-$. Since $A \cap \{[0]\} = \varnothing$, it follows that $A \subset W$ and, hence, that $A \in W^+$. Therefore, $A \in U$. This implies that U is an open neighborhood of A in $(\mathrm{CL}_{\neq\varnothing}(\mathbb{S}^1), \mathcal{T}_\mathcal{F})$.

The next result characterizes complete metrizability of the Fell topology on $\mathrm{CL}_{\neq\varnothing}(X)$.

Proposition 3.5 (Metrizability of the Fell Topology, [12, Prop. 5.1.5])**.** *Let $(X, \mathcal{T})$ be a Hausdorff space and $\mathcal{T}_\mathcal{F}$ be the Fell topology on $\mathrm{CL}_{\neq\varnothing}(X)$. The following are equivalent:*

- *$(X, \mathcal{T})$ is locally compact and second countable;*
- *$(\mathrm{CL}_{\neq\varnothing}(X), \mathcal{T}_\mathcal{F})$ is a separable and completely metrizable Hausdorff space.*

The result in Proposition 3.5 yields the following corollary via [12, Thm. 5.2.6].

Corollary 3.6 (Fell Topology and Painlevé-Kuratowski Convergence)**.** *Let $(X, \mathcal{T})$ be a locally compact and second countable Hausdorff space, $\{A_k\}_{k\in\mathbb{N}} \subset \mathrm{CL}_{\neq\varnothing}(X)$, and $A \in \mathrm{CL}_{\neq\varnothing}(X)$. Then, $\{A_k\}_{k\in\mathbb{N}} \xrightarrow{\mathcal{T}_\mathcal{F}} A$ in the Fell topology $\mathcal{T}_\mathcal{F}$ if and only if $\{A_k\}_{k\in\mathbb{N}} \to A$ in the Painlevé-Kuratowski sense.*

Corollary 3.6 enables the identification of properties of set-valued maps formulated via Painlevé-Kuratowski convergence in X, such as outer semicontinuous dependence, with their topological counterparts in $\mathrm{CL}_{\neq\varnothing}(X)$; see Section V.

IV. Geometric Hybrid Dynamical Systems

We define geometric hybrid systems evolving on an n-dimensional C^1-manifold $\mathcal{M}$ via the hybrid inclusion

$$\mathcal{H} : \begin{cases} \dot{x} \in F(x) & x \in C \\ x^+ \in G(x) & x \in D, \end{cases} \tag{1}$$

where $x \in \mathcal{M}$ is the state, $C \subset \mathcal{M}$ and $D \subset \mathcal{M}$ are the flow set and the jump set, respectively, and $G : \mathcal{M} \rightrightarrows \mathcal{M}$ is the jump map. The flow map $F : \mathcal{M} \rightrightarrows T\mathcal{M}$ is map satisfying $F(x) \subset T_x\mathcal{M}$ for each $x \in C$. We write $\mathcal{H} = (C, F, D, G, \mathcal{M})$ to refer to a geometric hybrid system with this data.

Solutions to $\mathcal{H}$ are defined on *hybrid time domains*. A *hybrid time domain* is the union of an increasing sequence

[2]Given a topological space $(X, \mathcal{T})$, $\{x_k\}_{k\in\mathbb{N}} \subset X$ converges to $x \in X$ if, for each $U \in \mathcal{N}(x)$, there exists $k_0 \in \mathbb{N}$ such that $x_k \in U$ for all $k \geq k_0$.

[3]Given a topological space $(X, \mathcal{T})$ and an equivalence relation $\sim$ on X, the projection $\pi : X \to X/\!\sim$ mapping each $x \in X$ to its equivalence class $[x] := \{y \in X : y \sim x\}$ determines the *quotient topology* on $X/\!\sim$, where $U \subset X/\!\sim$ is open if and only if $\pi^{-1}(U) \in \mathcal{T}$.

of nested compact hybrid time domains. A set $E \subset \mathbb{R}_{\geq 0} \times \mathbb{N}$ is a *compact hybrid time domain* if $E = \bigcup_{j=0}^{J-1} [t_j, t_{j+1}] \times \{j\}$ for some $J \in \mathbb{N}$ and some times $\{t_j\}_{j=0}^{J} \subset \mathbb{R}_{\geq 0}$ satisfying $0 = t_0 \leq t_1 \leq \ldots \leq t_J$.

Definition 4.1 (Hybrid Arc)**.** *A function $\phi : E \to \mathcal{M}$ is a hybrid arc if E is a hybrid time domain and if, for each $j \in \mathbb{N}$, the function $t \mapsto \phi(t, j)$ is locally absolutely continuous*[4] *on the interval $I_\phi^j := \{t : (t, j) \in E\}$.*

Given a hybrid arc ϕ, its total time of flow is given by $\sup_t \operatorname{dom} \phi := \sup\{t : \exists j \text{ s.t. } (t, j) \in \operatorname{dom} \phi\}$, and the number of times it jumps is given by $\sup_j \operatorname{dom} \phi := \sup\{j : \exists t \text{ s.t. } (t, j) \in \operatorname{dom} \phi\}$. The length of its domain is defined as $\operatorname{length}(\operatorname{dom} \phi) := \sup_t \operatorname{dom} \phi + \sup_j \operatorname{dom} \phi$.

The truncation of a hybrid arc ϕ up to $\tau \in \mathbb{R}_{\geq 0}$ is the hybrid arc $\phi|_{\leq \tau} : \operatorname{dom} \phi|_{\leq \tau} \to \mathcal{M}$ with $\operatorname{dom} \phi|_{\leq \tau} := \{(t, j) \in \operatorname{dom} \phi : t + j \leq \tau\}$ and $\phi|_{\leq \tau}(t, j) := \phi(t, j)$ for all $(t, j) \in \operatorname{dom} \phi|_{\leq \tau}$. The arcs $\phi|_{<\tau}$ and $\phi|_{\geq \tau}$, are defined similarly.

Definition 4.2 (Solution to a Hybrid Dynamical System)**.** *A hybrid arc ϕ is a solution to the geometric hybrid system $\mathcal{H} = (C, F, D, G, \mathcal{M})$ if $\phi(0, 0) \in \overline{C} \cup D$ and the following conditions hold: for each $j \in \mathbb{N}$ such that I_ϕ^j has nonempty interior, ϕ satisfies $\phi(t, j) \in C$ for all $t \in \operatorname{int} I_\phi^j$ and $\frac{d\phi}{dt}(t, j) \in F(\phi(t, j))$ for almost all $t \in I_\phi^j$; for each $(t, j) \in \operatorname{dom} \phi$ such that $(t, j + 1) \in \operatorname{dom} \phi$, ϕ satisfies $\phi(t, j) \in D$ and $\phi(t, j + 1) \in G(\phi(t, j))$.*

A solution ϕ is nontrivial if $\operatorname{dom} \phi$ contains at least two points. It is maximal if no other solution ψ to $\mathcal{H}$ exists such that $\operatorname{dom} \phi$ is a proper subset of $\operatorname{dom} \psi$ and $\phi(t, j) = \psi(t, j)$ for all $(t, j) \in \operatorname{dom} \phi$. A solution ϕ is said to be complete if $\operatorname{length}(\operatorname{dom} \phi) = \infty$. It is precompact if $\operatorname{rge} \phi$ is precompact. A solution ϕ is said to exhibit finite-time blowup if $\operatorname{length}(\operatorname{dom} \phi) < \infty$ and for each compact set $K \subset \mathcal{M}$ there exists $(t', j') \in \operatorname{dom} \phi$ such that $\operatorname{rge}(\phi|_{\geq t' + j'}) \cap K = \varnothing$.

The set of maximal solutions to $\mathcal{H}$ from $K \subset \mathcal{M}$ is denoted by $\mathcal{S}_\mathcal{H}(K)$. The set of maximal solutions from $K \subset \mathcal{M}$ that evolve up to time τ is defined by $\mathcal{S}_\mathcal{H}^{\leq \tau}(K) := \{\phi|_{\leq \tau} : \phi \in \mathcal{S}_\mathcal{H}(K)\}$. When $K = \{\phi_0\}$ for $\phi_0 \in \mathcal{M}$, we write $\mathcal{S}_\mathcal{H}(\phi_0)$ instead of $\mathcal{S}_\mathcal{H}(\{\phi_0\})$. When no initial condition is specified, $\phi \in \mathcal{S}_\mathcal{H}$ denotes any maximal solution to $\mathcal{H}$.

The following example shows that maximality alone does not guarantee that a solution has closed graph.

Example 4.3 (Finite-Time Non-Singleton Accumulation)**.** *Consider $\mathcal{H} = (C, F, D, G, \mathcal{M})$ with $\mathcal{M} = \mathbb{R}^2$, $C = \{(s, x) \in \mathbb{R}^2 : s \leq 1\}$, $D = \varnothing$, and flow map $F : \mathbb{R}^2 \to \mathbb{R}^2$ given by $F(s, x) := \left(1, \cos\left(\frac{1}{1-s}\right) \frac{1}{(1-s)^2}\right)$ for $(s, x) \in \operatorname{int} C$ and $F(s, x) = 0$ for $(s, x) \in \{1\} \times \mathbb{R}$. The maximal solution from $(0, 0)$ is given by $\phi(t, 0) = \left(t, \sin\left(\frac{1}{1-t}\right)\right)$ for $t \in [0, 1)$. However, $\operatorname{gph} \phi$ is not closed since, for each $y \in [-1, 1]$, $(1, y)$ is an accumulation point of $\phi(t_k, 0)$ along some sequence $t_k \to 1^-$, yet $(1, 0, 1, y) \notin \operatorname{gph} \phi$.*

To circumvent the behavior presented in Example 4.3, which might preclude the identification of maximal solutions as points in $\mathrm{CL}_{\neq \varnothing}(\mathbb{R}_{\geq 0} \times \mathbb{N} \times \mathcal{M})$ as well as the application of the Fell topology machinery of Section III, and to obtain structural properties of solution sets to geometric hybrid systems, we use the notion of graphical convergence.

Definition 4.4 (Graphical Convergence)**.** *A sequence $\{\phi_k\}_{k \in \mathbb{N}}$ of solutions ϕ to $\mathcal{H}$ is said to converge graphically if the sequence of sets $\{\operatorname{gph} \phi_k\}_{k \in \mathbb{N}}$ converges in $\mathbb{R}_{\geq 0} \times \mathbb{N} \times \mathcal{M}$ in the sense of Definition 2.1. The graphical limit of this sequence of solutions is the (possibly) set-valued map $\phi : \mathbb{R}_{\geq 0} \times \mathbb{N} \rightrightarrows \mathcal{M}$ defined by $\operatorname{gph} \phi = \lim_{k \to \infty} \operatorname{gph} \phi_k$. We often use the notation $\operatorname{gph-lim}_{k \to \infty} \phi_k$ to refer to such map.*

Remark 4.5 (Graphical Convergence in the Euclidean Setting)**.** *In the hybrid inclusions framework in $\mathbb{R}^n$, graphical convergence of solutions [1, Def. 5.21] is defined as convergence of their graphs as subsets of $\mathbb{R}_{\geq 0} \times \mathbb{N} \times \mathbb{R}^n$ in the sense of [1, Def. 5.1]. This set convergence notion is equivalent to Painlevé-Kuratowski convergence in the sense of Definition 2.1. Hence, Definition 4.4 coincides with [1, Def. 5.21] on $\mathbb{R}^n$ and extends it to a C^1-manifold $\mathcal{M}$.*

In general, graphical limits of sequences of solutions to geometric hybrid systems need not correspond to a solution, or even to a single-valued map. Nominal well-posedness imposes conditions that rule out these pathological cases.

Definition 4.6 (Nominally Well-Posed Hybrid Systems [3, Def. 8])**.** *A geometric hybrid system $\mathcal{H} = (C, F, D, G, \mathcal{M})$ is said to be nominally well-posed if, for each graphically convergent sequence of solutions $\{\phi_k\}_{k \in \mathbb{N}}$ to $\mathcal{H}$ satisfying $\lim_{k \to \infty} \phi_k(0, 0) = \phi_0 \in \mathcal{M}$, one of the following holds:*

(i) If $\{\phi_k\}_{k \in \mathbb{N}}$ is locally eventually precompact,[5] *there exists $\tau \in \mathbb{R}_{\geq 0} \cup \{\infty\}$ such that $\lim_{k \to \infty} \operatorname{length}(\operatorname{dom} \phi_k) = \tau$ and $\phi := \operatorname{gph-lim}_{k \to \infty} \phi_k$ is such that ϕ is a solution to $\mathcal{H}$ with $\phi(0, 0) = \phi_0$ and $\operatorname{length}(\operatorname{dom} \phi) = \tau$;*

(ii) If $\{\phi_k\}_{k \in \mathbb{N}}$ is not locally eventually precompact, there exist $\tau \in \mathbb{R}_{\geq 0}$ and a sequence $\{(t_k, j_k)\}_{k \in \mathbb{N}} \subset \operatorname{dom} \phi_k$ with $\lim_{k \to \infty} t_k + j_k = \tau$ such that

- *(a) for each compact set $K \subset \mathcal{M}$, there exists $k_0 \in \mathbb{N}$ such that $\phi_k(t_k, j_k) \notin K$ for each $k > k_0$, and*
- *(b) the truncated limit $\phi := (\operatorname{gph-lim}_{k \to \infty} \phi_k)|_{<\tau}$ is such that $\phi \in \mathcal{S}_\mathcal{H}(\phi_0)$, $\operatorname{length}(\operatorname{dom} \phi) = \tau$, and, for each compact set $K \subset \mathcal{M}$, there exists $\tau' \in (0, \tau)$ such that $K \cap \operatorname{rge} \phi|_{\geq \tau'} = \varnothing$.*

Mild conditions on the data of a geometric hybrid system $\mathcal{H}$, known as the hybrid basic conditions [1, Assmp. 6.5], have been shown to guarantee nominal well-posedness in the case $\mathcal{M} = \mathbb{R}^n$ [1, Thm. 6.8]. When $\mathcal{M}$ is a C^1-manifold, similar conditions ensure that case (i) of Definition 4.6 holds for every convergent sequence of solutions that is locally eventually precompact [3, Thm. 2].

[4] Given a C^k-manifold $\mathcal{M}$ and an interval $I \subset \mathbb{R}$, a function $f : I \to \mathcal{M}$ is locally absolutely continuous if $\varphi \circ f$ is locally absolutely continuous for every k-times continuously differentiable $\varphi : \mathcal{M} \to \mathbb{R}$ [13, Sec. A.2.1].

[5] A sequence of solutions $\{\phi_k\}_{k \in \mathbb{N}}$ to $\mathcal{H}$ is *locally eventually precompact* if, for each $\tau \in \mathbb{R}_{\geq 0}$, there exists $k_0 \in \mathbb{N}$ such that $\operatorname{rge} \phi_k|_{<\tau}$ is precompact for each $k > k_0$.

V. Topological Properties of Solution Sets

In this section, we establish topological properties of the solution map of nominally well-posed hybrid systems. First, we show that maximal solutions have closed graph.

Proposition 5.1 (Maximal Solutions and Closed Graph)**.** *Let $\mathcal{H} = (C, F, D, G, \mathcal{M})$ be a nominally well-posed geometric hybrid system. Then, for each $\phi \in \mathcal{S}_{\mathcal{H}}$, $\operatorname{gph} \phi$ is closed.*

Proposition 5.1 allows each maximal solution to be identified with a point in $\mathrm{CL}_{\neq\varnothing}(\mathbb{R}_{\geq 0} \times \mathbb{N} \times \mathcal{M})$. We encode this identification via the map introduced next.

Definition 5.2 (Solution Map)**.** *Given a nominally well-posed geometric hybrid system $\mathcal{H} = (C, F, D, G, \mathcal{M})$, the solution map on the hyperspace $\mathcal{G}_{\mathcal{H}} : \mathbb{R}_{\geq 0} \times \mathcal{M} \rightrightarrows \mathrm{CL}_{\neq\varnothing}(\mathbb{R}_{\geq 0} \times \mathbb{N} \times \mathcal{M})$ is defined, for each $(\tau, \phi_0) \in \mathbb{R}_{\geq 0} \times \mathcal{M}$, by*

$$\mathcal{G}_{\mathcal{H}}(\tau, \phi_0) \coloneqq \{\operatorname{gph} \phi : \phi \in \mathcal{S}_{\mathcal{H}}^{\leq \tau}(\phi_0)\}.$$

To obtain suitable regularity properties of $\mathcal{G}_{\mathcal{H}}$ under nominal well-posedness of a geometric hybrid system $\mathcal{H}$, we leverage the notion of pre-forward completeness.

Definition 5.3 (Pre-forward Completeness)**.** *Given a set $\Lambda \subset \mathcal{M}$, the geometric hybrid system $\mathcal{H}$ is said to be pre-forward complete from Λ if every maximal solution $\phi \in \mathcal{S}_{\mathcal{H}}(\Lambda)$ is either precompact or complete.*

Pre-forward completeness of a nominally well-posed geometric hybrid system from a set rules out finite-time blowup of solutions and ensures that graphically convergent sequences of solutions are locally eventually precompact.

Lemma 5.4 (Pre-forward Completeness and Local Eventual Precompactness)**.** *Let $\mathcal{H} = (C, F, D, G, \mathcal{M})$ be a nominally well-posed geometric hybrid system that is pre-forward complete from a set $\Lambda \subset \mathcal{M}$. Then, every graphically convergent sequence of solutions $\{\phi_k\}_{k\in\mathbb{N}}$ to $\mathcal{H}$ with $\lim_{k\to\infty} \phi_k(0, 0) = \phi_0 \in \Lambda$ is locally eventually precompact.*

Next, since $\mathcal{M}$ is locally compact and second countable, $(\mathrm{CL}_{\neq\varnothing}(\mathbb{R}_{\geq 0} \times \mathbb{N} \times \mathcal{M}), \mathcal{T}_{\mathcal{F}})$ is metrizable and, thus, outer semicontinuity of $\mathcal{G}_{\mathcal{H}}$ can be characterized along every convergent sequence $(\tau_k, \phi_{0,k}) \to (\tau_0, \phi_0)$, as in Definition 2.2. We leverage this fact together with Lemma 5.4 to identify each element of the outer limit along such a sequence as an element of $\mathcal{G}_{\mathcal{H}}(\tau_0, \phi_0)$. In $\mathbb{R}^n$, the same dependence on initial conditions is stated through a compact set K, its inflations $K + \delta\mathbb{B}$, and (τ, ε)-closeness [1, Prop. 6.14], whereas here the converging solutions are already part of the definition.

Theorem 5.5 (Outer Semicontinuity of the Solution Map)**.** *Let $\mathcal{H} = (C, F, D, G, \mathcal{M})$ be a nominally well-posed geometric hybrid system and suppose that $\mathcal{H}$ is pre-forward complete from a set $\Lambda \subset \mathcal{M}$. Then, the solution map $\mathcal{G}_{\mathcal{H}} : \mathbb{R}_{\geq 0} \times \mathcal{M} \rightrightarrows \mathrm{CL}_{\neq\varnothing}(\mathbb{R}_{\geq 0} \times \mathbb{N} \times \mathcal{M})$ is outer semicontinuous at each $(\tau_0, \phi_0) \in \mathbb{R}_{\geq 0} \times \Lambda$ with respect to the Fell topology $\mathcal{T}_{\mathcal{F}}$ on $\mathrm{CL}_{\neq\varnothing}(\mathbb{R}_{\geq 0} \times \mathbb{N} \times \mathcal{M})$.*

Now, by using Lemma 5.4 and the fact that a C^1-manifold $\mathcal{M}$ is Hausdorff and locally compact, hence regular[6], we obtain that $\mathcal{G}_{\mathcal{H}}$ is locally precompact.

Theorem 5.6 (Local Precompactness of the Solution Map)**.** *Let $\mathcal{H} = (C, F, D, G, \mathcal{M})$ be a nominally well-posed geometric hybrid system and suppose that $\mathcal{H}$ is pre-forward complete from a neighborhood of a set $\Lambda \subset \mathcal{M}$. Then, for each $(\tau_0, \phi_0) \in \mathbb{R}_{\geq 0} \times \Lambda$, the solution map $\mathcal{G}_{\mathcal{H}}$ is locally precompact at (τ_0, ϕ_0) with respect to the Fell topology $\mathcal{T}_{\mathcal{F}}$ on $\mathrm{CL}_{\neq\varnothing}(\mathbb{R}_{\geq 0} \times \mathbb{N} \times \mathcal{M})$.*

Next, using Theorems 5.5 and 5.6, we obtain that the values of $\mathcal{G}_{\mathcal{H}}$ in $(\mathrm{CL}_{\neq\varnothing}(\mathbb{R}_{\geq 0} \times \mathbb{N} \times \mathcal{M}), \mathcal{T}_{\mathcal{F}})$ are compact.

Corollary 5.7 (Compactness of the Set of Solutions)**.** *Let $\mathcal{H} = (C, F, D, G, \mathcal{M})$ be a nominally well-posed geometric hybrid system and suppose that $\mathcal{H}$ is pre-forward complete from a neighborhood of a set $\Lambda \subset \mathcal{M}$. Then, $\mathcal{G}_{\mathcal{H}}(\tau_0, \phi_0)$ is compact in $(\mathrm{CL}_{\neq\varnothing}(\mathbb{R}_{\geq 0} \times \mathbb{N} \times \mathcal{M}), \mathcal{T}_{\mathcal{F}})$ for each $(\tau_0, \phi_0) \in \mathbb{R}_{\geq 0} \times \Lambda$.*

The following corollary establishes measurability of $\mathcal{G}_{\mathcal{H}}|_{\mathbb{R}_{\geq 0}\times\Lambda}$ from outer semicontinuity of $\mathcal{G}_{\mathcal{H}}$, together with σ-compactness[7] of both $\mathbb{R}_{\geq 0} \times \mathcal{M}$ and $(\mathrm{CL}_{\neq\varnothing}(\mathbb{R}_{\geq 0} \times \mathbb{N} \times \mathcal{M}), \mathcal{T}_{\mathcal{F}})$, which enables the application of [5, Thm. 18.20].

Corollary 5.8 (Measurability of the Solution Map)**.** *Let $\mathcal{H} = (C, F, D, G, \mathcal{M})$ be a nominally well-posed geometric hybrid system and $\Lambda \subset \mathcal{M}$ be closed. Suppose $\mathcal{H}$ is pre-forward complete from a neighborhood of Λ. Then, the restricted solution map $\mathcal{G}_{\mathcal{H}}|_{\mathbb{R}_{\geq 0}\times\Lambda} : \mathbb{R}_{\geq 0} \times \mathcal{M} \rightrightarrows \mathrm{CL}_{\neq\varnothing}(\mathbb{R}_{\geq 0} \times \mathbb{N} \times \mathcal{M})$ is compact valued and is a measurable set-valued map, in the sense of Definition 2.4, from the measurable space $(\mathbb{R}_{\geq 0} \times \mathcal{M}, \mathfrak{B}(\mathbb{R}_{\geq 0} \times \mathcal{M}))$ to the separable metrizable space $(\mathrm{CL}_{\neq\varnothing}(\mathbb{R}_{\geq 0} \times \mathbb{N} \times \mathcal{M}), \mathcal{T}_{\mathcal{F}})$.*

Remark 5.9 (Sequential and topological properties)**.** *In $\mathbb{R}^n$, outer semicontinuous dependence of solutions on initial conditions and sequential compactness of $\mathcal{S}_{\mathcal{H}}^{\leq\tau}(K)$, for a compact $K \subset \mathbb{R}^n$, are established through graphical convergence in [1, Prop. 6.14, Thm. 6.8]; see Remark 4.5. In Theorems 5.5 and 5.6, we extend these results to C^1-manifolds and obtain outer semicontinuity and local precompactness of $\mathcal{G}_{\mathcal{H}}$ in $(\mathrm{CL}_{\neq\varnothing}(\mathbb{R}_{\geq 0} \times \mathbb{N} \times \mathcal{M}), \mathcal{T}_{\mathcal{F}})$. To prove these theorems, we extend to $\mathcal{M}$ the sequential tools of the Euclidean setting, namely, extraction of graphically convergent subsequences and nominal well-posedness, and combine them with the geometry of $\mathcal{M}$ and the structure of the hyperspace $(\mathrm{CL}_{\neq\varnothing}(\mathbb{R}_{\geq 0} \times \mathbb{N} \times \mathcal{M}), \mathcal{T}_{\mathcal{F}})$. In Section VI, we identify the domain and codomain spaces of $\mathcal{G}_{\mathcal{H}}$ and their regularity and, together with outer semicontinuity and measurability, verify the assumptions of general results from the set-valued analysis literature (e.g., [5, Thm. 18.19, Lem. 17.30,]), which allows us to obtain existence of extremal solutions for a given cost.*

[6] A topological space $(X, \mathcal{T})$ is *regular* if, for each $x \in X$ and each closed set $A \subset X$ with $x \notin A$, there exist disjoint open sets $U, V \in \mathcal{T}$ with $x \in U$ and $A \subset V$.

[7] A topological space is *σ-compact* if it can be written as a countable union of compact subsets.

VI. Application: Existence of Extremal Solutions

We apply the topological properties of $\mathcal{G}_{\mathcal{H}}$ from Section V to obtain optimal solutions of cost functionals on solution sets and measurable selections of optimizers under random initial conditions. Related results in $\mathbb{R}^n$, not following a topological approach in the hyperspace, can be found in [14].

First, using the outer semicontinuity and local precompactness of the solution map $\mathcal{G}_{\mathcal{H}}$, we apply [5, Lem. 17.30] to obtain existence of extremal solutions for a class of cost functionals defined on solution graphs.

Proposition 6.1 (Optimization of Costs over Solution Sets)**.** *Let $\mathcal{H} = (C, F, D, G, \mathcal{M})$ be a nominally well-posed geometric hybrid system and suppose $\mathcal{H}$ is pre-forward complete from a neighborhood of $\Lambda \subset \overline{C} \cup D$. Let $\mathcal{J} : \operatorname{gph} \mathcal{G}_{\mathcal{H}}|_{\mathbb{R}_{\geq 0} \times \Lambda} \to \mathbb{R}$ be upper semicontinuous[8], where $\operatorname{gph} \mathcal{G}_{\mathcal{H}}|_{\mathbb{R}_{\geq 0} \times \Lambda}$ is equipped with the subspace topology. Then, for each $(\tau_0, \phi_0) \in \mathbb{R}_{\geq 0} \times \Lambda$, there exists $\phi^\star \in \mathcal{S}_{\mathcal{H}}^{\leq \tau_0}(\phi_0)$ with*

$$\mathcal{J}(\tau_0, \phi_0, \operatorname{gph} \phi^\star) = \max_{\tilde{\phi} \in \mathcal{G}_{\mathcal{H}}(\tau_0, \phi_0)} \mathcal{J}(\tau_0, \phi_0, \tilde{\phi}), \tag{2}$$

and the value function $m : \mathbb{R}_{\geq 0} \times \Lambda \to \mathbb{R}$ defined by $m(\tau_0, \phi_0) := \mathcal{J}(\tau_0, \phi_0, \operatorname{gph} \phi^\star)$ is upper semicontinuous.

Example 6.2. *Let $\mathcal{H} = (C, F, D, G, \mathcal{M})$ be a nominally well-posed geometric hybrid system, $\Lambda \subset \overline{C} \cup D$, and suppose $\mathcal{H}$ is pre-forward complete from a neighborhood of Λ. Let $c : \mathcal{M} \to \mathbb{R}$ be upper semicontinuous. Define $\mathcal{J} : \operatorname{gph} \mathcal{G}_{\mathcal{H}}|_{\mathbb{R}_{\geq 0} \times \Lambda} \to \mathbb{R}$, for each $(\tau_0, \phi_0) \in \mathbb{R}_{\geq 0} \times \Lambda$ and each $\operatorname{gph} \phi \in \mathcal{G}_{\mathcal{H}}(\tau_0, \phi_0)$, by*

$$\mathcal{J}(\tau_0, \phi_0, \operatorname{gph} \phi) := \sum_{j=0}^{\sup_j \operatorname{dom} \phi - 1} c(\phi(t_{j+1}, j)), \tag{3}$$

where $t_{j+1} = \sup I_\phi^j$ and I_ϕ^j is as in Definition 4.1. Then, $\mathcal{J}$ is upper semicontinuous in the sense of Proposition 6.1.

Therefore, for each $(\tau_0, \phi_0) \in \mathbb{R}_{\geq 0} \times \Lambda$, there exists $\phi^\star \in \mathcal{S}_{\mathcal{H}}^{\leq \tau_0}(\phi_0)$ such that $\mathcal{J}(\tau_0, \phi_0, \operatorname{gph} \phi^\star) = \max_{\tilde{\phi} \in \mathcal{G}_{\mathcal{H}}(\tau_0, \phi_0)} \mathcal{J}(\tau_0, \phi_0, \tilde{\phi})$. Moreover, the value function $m : \mathbb{R}_{\geq 0} \times \Lambda \to \mathbb{R}$ defined by $m(\tau_0, \phi_0) := \mathcal{J}(\tau_0, \phi_0, \operatorname{gph} \phi^\star)$ is upper semicontinuous.

Now, when ϕ_0 is a measurable map from a measurable space $(\Omega, \mathfrak{A})$ to $\mathcal{M}$, using Corollary 5.8, we obtain a measurable value function and a measurable selection of optimizers by using the measurable maximum theorem in [5, Thm. 18.19].

Proposition 6.3 (Measurability under Random Initial Conditions)**.** *Let $\mathcal{H} = (C, F, D, G, \mathcal{M})$ be a nominally well-posed geometric hybrid system and $(\Omega, \mathfrak{A})$ be a measurable space. Suppose $\mathcal{H}$ is pre-forward complete from a neighborhood of a closed set $\Lambda \subset \overline{C} \cup D$. Let $\omega \mapsto \phi_0(\omega) \in \Lambda$ be a measurable function, fix $\tau_0 > 0$, and suppose $\mathcal{J} : \Omega \times \mathrm{CL}_{\neq \varnothing}(\mathbb{R}_{\geq 0} \times \mathbb{N} \times \mathcal{M}) \to \mathbb{R}$ is a Carathéodory map[9]. Then, for each $\omega \in \Omega$, there exists $\phi^\star(\omega) \in \mathcal{S}_{\mathcal{H}}^{\leq \tau_0}(\phi_0(\omega))$ such that*

$$\mathcal{J}(\omega, \operatorname{gph} \phi^\star(\omega)) = \max_{\tilde{\phi} \in \mathcal{G}_{\mathcal{H}}(\tau_0, \phi_0(\omega))} \mathcal{J}(\omega, \tilde{\phi}). \tag{4}$$

Moreover, the value function $m : \Omega \to \mathbb{R}$ defined by $m(\omega) := \mathcal{J}(\omega, \operatorname{gph} \phi^\star(\omega))$ is $\mathfrak{A}$-measurable, the argmax map $\mu(\omega) = \{\tilde{\phi} \in \mathcal{G}_{\mathcal{H}}(\tau_0, \phi_0(\omega)) : \mathcal{J}(\omega, \tilde{\phi}) = m(\omega)\}$ has nonempty compact values and is $\mathfrak{A}$-measurable, and there exists a $\mathfrak{A}$-measurable selector $\omega \mapsto \operatorname{gph} \phi^\star(\omega) \in \mu(\omega)$.

VII. Conclusion

This paper developed a topological framework for solution sets of geometric hybrid systems defined on a C^1-manifold $\mathcal{M}$, establishing outer semicontinuity, local precompactness, and measurability of the solution map with respect to the Fell topology on the hyperspace of closed and nonempty subsets of $\mathbb{R}_{\geq 0} \times \mathbb{N} \times \mathcal{M}$. These properties were applied to obtain existence of extremal solutions and measurable selection of optimizers under measurable choices of initial conditions. Future directions include extending the cost functionals over flow intervals and applying similar topological concepts in the context of hybrid optimal control on manifolds.

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[8] A function $f : Z \to \mathbb{R}$ on a topological space Z is *upper semicontinuous* if $\{z \in Z : f(z) < \alpha\}$ is open for every $\alpha \in \mathbb{R}$.

[9] A map $f : \Omega \times Y \to \mathbb{R}$ is said to be a *Carathéodory* map if $\omega \mapsto f(\omega, y)$ is $\mathfrak{A}$-measurable for each $y \in Y$ and $y \mapsto f(\omega, y)$ is continuous for each $\omega \in \Omega$.